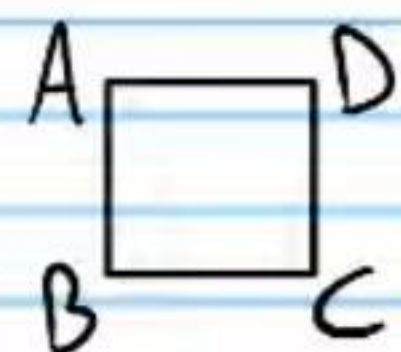


Special Quadrilaterals (Part III): More Proofs

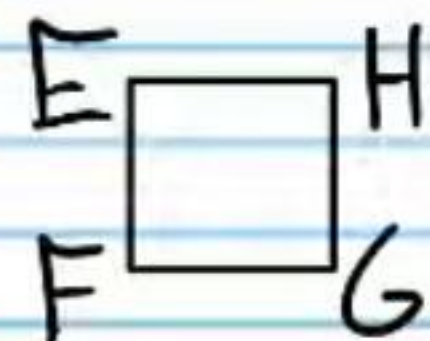
Write a two-column proof for each statement below.

① If the quadrilateral ABCD below is a square, then it is a rectangle (this is the Square Rectangle Theorem).



Statement	Reason
I) ABCD is a square.	Given.
II) $\angle A, \angle B, \angle C, \& \angle D$ are rt $\angle$'s.	Def of a square.
III) ABCD is a rectangle.	Def of a rectangle.

② If the quadrilateral EFGH below is a square, then it is a rhombus (this is the Square Rhombus Theorem).



Statement	Reason
I) EFGH is a square.	Given.
II) $\overline{EF} \cong \overline{FG} \cong \overline{GH} \cong \overline{EH}$	Def. of a square.
III) EFGH is a rhombus.	Def of a rhombus.

③ If the quadrilateral IJKL below is a rhombus, then its diagonals bisect the inner angles (this is the Rhombus Angle Bisector Theorem).



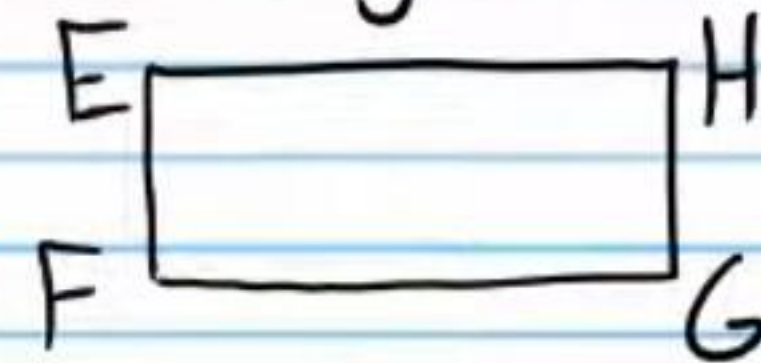
Statement	Reason
I) IJKL is a rhombus	Given.
II) $\overline{IJ} \cong \overline{JK} \cong \overline{KL} \cong \overline{LI}$.	Def. of a rhombus.
III) $\overline{IK} \cong \overline{IK}$.	Ref. Prop of $\cong$.
IV) $\triangle JIK \cong \triangle LIK$	SSS
V) $\angle JIK \cong \angle LIK$ and $\angle JKI \cong \angle LKI$.	CPCTC

④ If the quadrilateral ABCD below is a kite, then the angles between congruent sides are bisected by the diagonal connecting the vertices of those angles (this is the Kite Angle Bisector Theorem).



Statement	Reason
I) ABCD is a kite.	Given.
II) $\overline{AD} \cong \overline{DC}$ & $\overline{AB} \cong \overline{BC}$.	Def of a kite.
III) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$.
IV) $\triangle ABD \cong \triangle CBD$	SSS
V) $\angle ADB \cong \angle CDB$ & $\angle ABD = \angle CBD$.	CPCTC

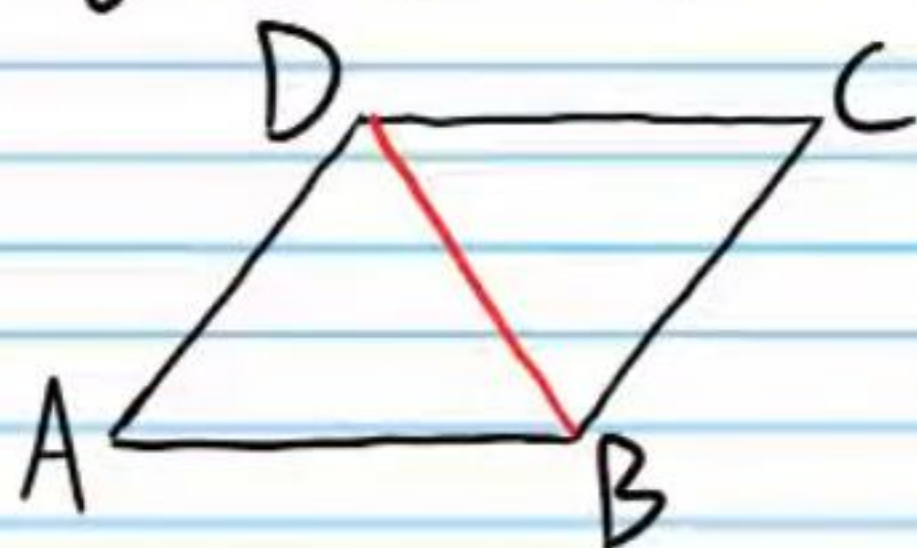
⑤ If the quadrilateral EFGH below is a rectangle, then it is a parallelogram (this is the Rectangle Parallelogram Theorem).



Statement	Reason
I) EFGH is a rectangle.	Given.
II) $\angle E$, $\angle F$, & $\angle H$ are rt. $\angle$'s.	Def of a rectangle
III) $m\angle E = 90^\circ$, $m\angle F = 90^\circ$, and $m\angle H = 90^\circ$.	Def of rt $\angle$.
IV) $m\angle E + m\angle F = 90 + 90$ and $m\angle E + m\angle H = 90 + 90$	Subs. Prop of $=$.
V) $m\angle E + m\angle F = 180$ and $m\angle E + m\angle H = 180$	Simplification.
VI) $\angle E$ & $\angle F$ are supplementary and $\angle E$ and $\angle H$ are supplementary	Def of supp. $\angle$'s.
VII) $\overline{EF} \parallel \overline{HG}$ & $\overline{EH} \parallel \overline{FG}$	CIACT
VIII) EFGH is a parallelogram.	Def of a $\square$.

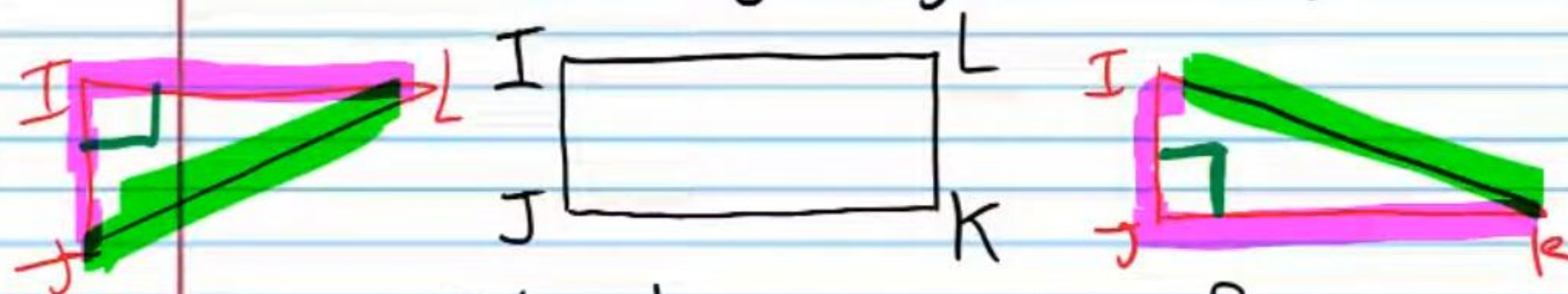
Consecutive Interior Angles Converse Theorem: If consecutive interior angles formed by a transversal are supplementary, then the lines cut by the transversal are parallel.

⑥ If the quadrilateral ABCD below is a rhombus, then it is a parallelogram (this is the Rhombus Parallelogram Theorem).



Statement	Reason
I) ABCD is a rhombus.	Given.
II) $\overline{AD} \cong \overline{BC}$ & $\overline{DC} \cong \overline{AB}$	Def of a rhombus
III) $\overline{BD} \cong \overline{BD}$	Ref. Prop. of $\cong$.
IV) $\triangle ADB \cong \triangle CBD$	SSS
V) $\angle ADB \cong \angle CBD$ & $\angle CDB \cong \angle DBA$	CPCTC.
VI) $\overline{DC} \parallel \overline{AB}$ & $\overline{AD} \parallel \overline{BC}$	Alt Angles
VII) ABCD is a parallelogram.	Def. of a $\square$.

⑦ If the quadrilateral IJKL below is a rectangle, then its diagonals are congruent (this is the Rectangle Diagonals Theorem).



Statement	Reason
I) IJKL is a rectangle.	Given.
II) $\overline{IJ} \cong \overline{IJ}$	Ref. Prop. of $\cong$.
III) IJKL is a parallelogram.	$\square$ $\square$ Thm.
IV) $\overline{IL} \cong \overline{JK}$	$\square$ Opp sides Thm.
V) $\angle I$ & $\angle J$ are rt $\angle$'s.	Def of a rectangle.
VI) $\triangle IJL \cong \triangle JIK$	SAS
VII) $\overline{IK} \cong \overline{JL}$	CPCTC

- ⑧ If the quadrilateral EFGH below is a rhombus, then its diagonals are perpendicular (this is the Rhombus Perpendicular Diagonals Theorem).

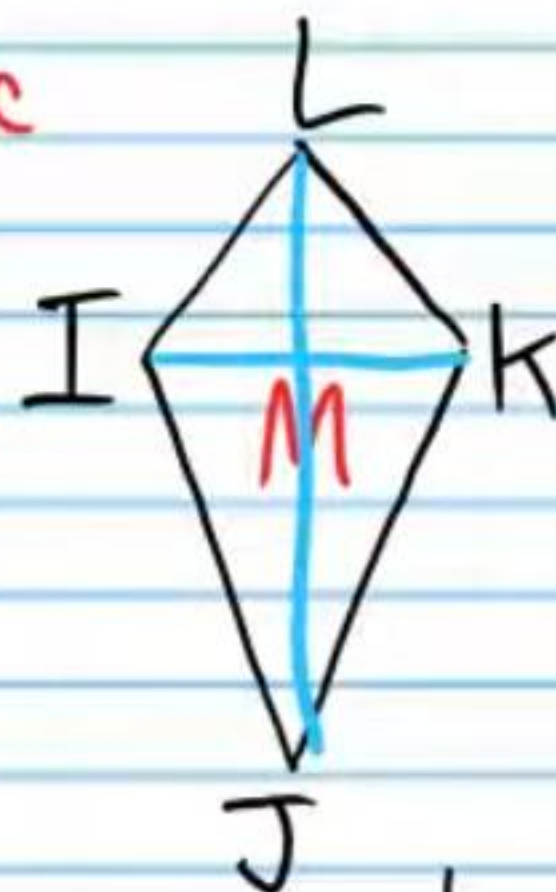


Statement	Reason
I) EFGH is a rhombus.	Given.
II) $\overline{EH} \cong \overline{HG}$	Def of a rhombus.
III) $\angle IHG \cong \angle EHI$	Rhombus $\angle$ Bisector Thm
IV) $\overline{HI} \cong \overline{HI}$	Ref. Prop of $\cong$.
V) $\triangle EHI \cong \triangle GHI$	SAS
VI) $\angle EIH \cong \angle GIH$	CPCTC
VII) $\angle EIH$ & $\angle GIH$ form a linear pair.	Def of a linear pair.
VIII) $\angle EIH$ & $\angle GIH$ are supp.	Linear Pair Post.
IX) $m\angle EIH + m\angle GIH = 180^\circ$	Def of Supp. $\angle$'s.
X) $m\angle EIH = m\angle GIH$	Def of $\cong \angle$'s.
XI) $m\angle EIH + m\angle EIH = 180^\circ$	Subs. Prop of $=$.

XII) $2m\angle EIH = 180$	Simplification.
XIII) $m\angle EIH = 90$	Div. Prop of $=$.
XIV) $\angle EIH$ is a rt $\angle$.	Def of a rt $\angle$.
XV) $\overline{EG} \perp \overline{FH}$	Def of $\perp$.

⑨ If the quadrilateral IJKL below is a kite, then its diagonals are perpendicular (this is the Kite Perpendicular Diagonals Theorem).

★ Use the Kite Angle Bisector Theorem and the Linear Pair Postulate. Use the letter M for the intersection point.



Statement	Reason
I) IJKL is a kite	Given.
II) $\overline{IL} \cong \overline{IK}$	Def of a kite.
III) $\overline{LM} \cong \overline{LM}$	Ref. Prop of $\cong$.
IV) $\angle ILM \cong \angle KLM$	kite $\angle$ Bisector Thm.
V) $\triangle ILM \cong \triangle KLM$	SAS
VI) $\angle IML \cong \angle KML$	CPCTC
VII) $m\angle IML = m\angle KML$	Def of $\cong \angle$'s.
VIII) $\angle IML$ & $\angle KML$ form a linear pair.	Def of a linear pair.
IX) $\angle IML$ & $\angle KML$ are supp. $\angle$'s.	Linear Pair Post.

X) $m\angle IML + m\angle KML = 180^\circ$	Def of Supp. $\angle$'s
XI) $m\angle IML + m\angle IML = 180^\circ$	Subs. Prop of $=$.
XII) $2m\angle IML = 180$	Simplification
XIII) $m\angle IML = 90$	Div. Prop of $=$.
XIV) $\overline{IJ} \perp \overline{IK}$	Def of $\perp$

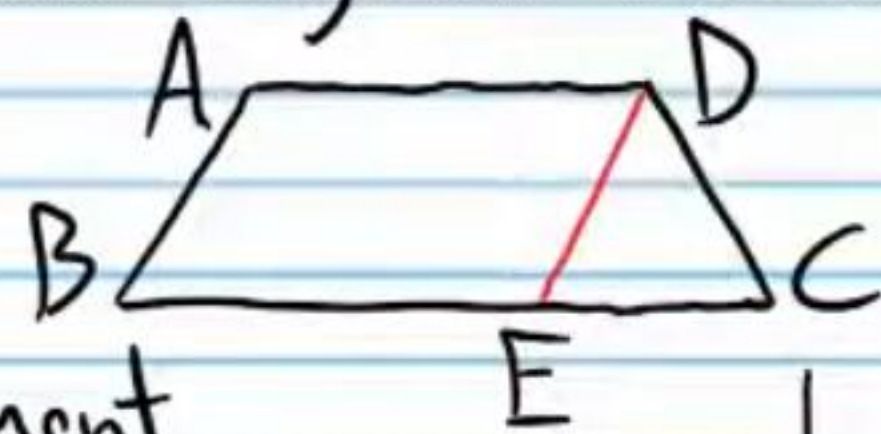
⑩ If the quadrilateral ABCD below is a kite, then its diagonals divide it into two pairs of congruent right triangles (this is the Kite Right Triangles Theorem).

★ You may use the Kite Perpendicular Diagonals Theorem and the Kite Diagonal Bisector Theorem. Use the letter E for the intersection point.



Statement	Reason
I) ABCD is a kite	Given.
II) $\overline{DB} \perp \overline{AC}$	Kite $\perp$ Diagonals Thm.
III) $\angle AED, \angle CED, \angle AEB,$ and $\angle CEB$ are rt $\angle$'s.	Def of $\perp$.
IV) $\angle AED \cong \angle CED$ and $\angle AEB \cong \angle CEB$.	Rt $\angle \cong$ Thm.
V) $\overline{DE} \cong \overline{DE}$ & $\overline{BE} \cong \overline{BE}$	Ref. Prop of $\cong$.
VI) $\overline{DB}$ bisects $\overline{AC}$	Kite Diagonal Bisector Thm.
VII) $\overline{AE} \cong \overline{EC}$	Def of seg bisector.
VIII) $\triangle AED \cong \triangle CED$ and $\triangle AEB \cong \triangle CEB$.	SAS

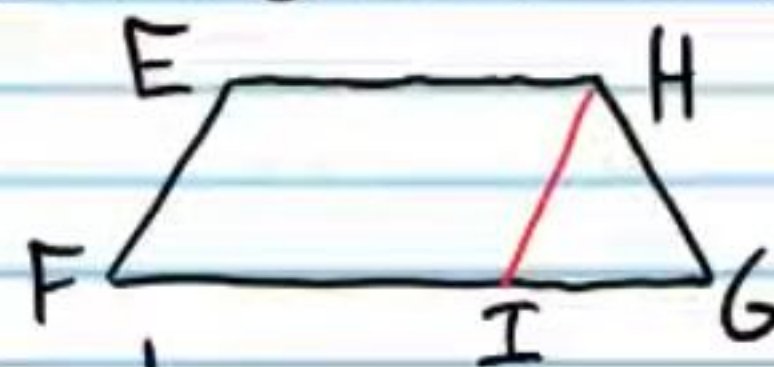
⑪ If the trapezoid ABCD below is isosceles, then the angles in each base pair are congruent (this is the Trapezoid Congruent Base Angles Theorem).



Statement	Reason
I) ABCD is an isosceles $\square$.	Given.
II) $\overline{AD} \parallel \overline{BC}$.	Def of a $\square$.
III) $\overline{AB} \parallel \overline{DE}$	Drawn.
IV) ABED is a parallelogram.	Def of a $\square$.
V) $\overline{AB} \cong \overline{DE}$	$\square$ Opp Sides Thm.
VI) $\overline{AB} \cong \overline{DC}$	Def of an isosceles $\square$.
VII) $\overline{DC} \cong \overline{DE}$	Trans. Prop of $\cong$.
VIII) $\angle DEC \cong \angle C$	Isosceles $\triangle$ Thm.
IX) $\angle B \cong \angle DEC$	CAP
X) $\angle B \cong \angle C$	Trans. Prop of $\cong$.
XI) $\angle A$ & $\angle B$ are supplementary and $\angle D$ & $\angle C$ are supplementary.	CIAT
XII) $m\angle A + m\angle B = 180$ and $m\angle D + m\angle C = 180$	Def of supp $\angle$'s.

- XIII) $m\angle A + m\angle B = m\angle D + m\angle C$ Trans. Prop of $=$
 XIV) $m\angle B = m\angle C$ DH of $\cong \angle$'s.
 XV) $m\angle A + m\angle C = m\angle D + m\angle C$ Subs. Prop of $=$.
 XVI) $m\angle A = m\angle D$ Subst. Prop of $=$.
 XVII) $\angle A \cong \angle D$ Def of $\cong \angle$'s.

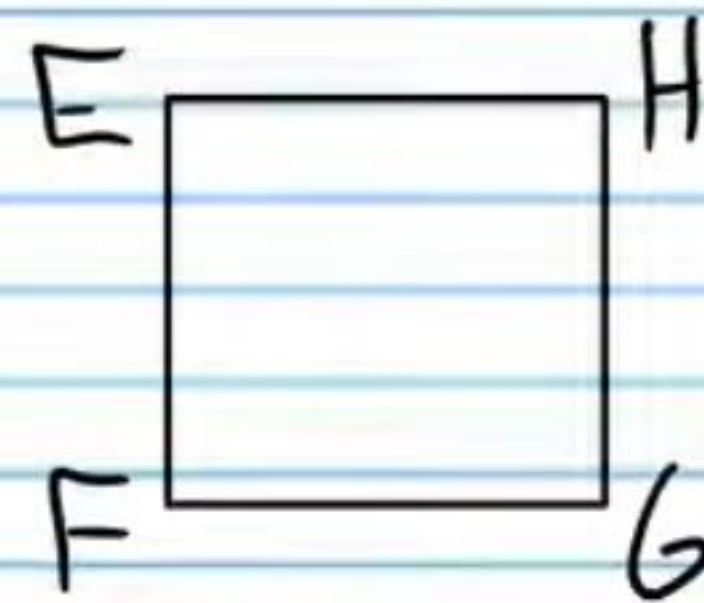
⑫ If $\angle F \cong \angle G$ in the trapezoid EFGH below, then it is an isosceles trapezoid (this is the Trapezoid One Pair of Base Angles Theorem).



Statement	Reason
I) EFGH is a trapezoid	Given.
II) $\overline{EH} \parallel \overline{FG}$	Def of a trapezoid.
III) $\overline{EF} \parallel \overline{HI}$	Drawn.
IV) EFHI is a parallelogram.	Def of a $\square$.
V) $\angle F \cong \angle HIG$	CAP
VI) $\angle F \cong \angle G$	Given
VII) $\angle HIG \cong \angle G$	Trans. Prop of $=$.
VIII) $\overline{HI} \cong \overline{HG}$	Isosceles Δ Conv Thm.

- IX) $\overline{FE} \cong \overline{HI}$ $\square$ Opp Sides Thm.
 X) $\overline{HG} \cong \overline{FE}$ Trans. Prop of $\cong$.
 XI) EFGH is an isosceles $\square$. Def of an isosceles $\square$.

⑬ If both diagonals are drawn in the square EFGH below, then four 45° - 45° - 90° triangles are created (this is the Square Four Special Triangles Theorem).



Statement	Reason
I)	
II)	
III)	
IV)	
V)	
VI)	